

MD. AKASH MIAH

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PROFESSIONAL SUMMARY

AI Engineer with experience designing, developing, and deploying production-grade Generative AI systems, LLM applications, RAG pipelines, and AI agents. Skilled in Python, FastAPI, vector databases, and modern AI frameworks including LangChain. Experienced in building scalable AI platforms, multi-model inference systems, intelligent automation solutions, and developer tooling that improve productivity and business workflows. Strong foundation in machine learning, deep learning, and software engineering with hands-on experience deploying AI systems from research to production.

TECHNICAL SKILLS

Programming	Python, Go, SQL, TypeScript
Generative AI	LLMs, RAG, AI Agents, Prompt Engineering, Function Calling, Model Evaluation
LLM Frameworks	LangChain, LangGraph, LlamaIndex
LLM Providers	OpenAI, Claude, Gemini, Groq, Self-Hosted Models
Vector Databases	Qdrant, Pinecone, PGVector, ChromaDB
Machine Learning	PyTorch, TensorFlow, Scikit-learn, Transformers, Fine-tuning
Backend & Deployment	FastAPI, Docker, REST APIs, PostgreSQL, Redis, JWT Authentication
Infrastructure	Linux, Git, vLLM, Model Serving, Prometheus, CI/CD
AI Domains	NLP, Computer Vision, Conversational AI, Semantic Search, Multi-Agent Systems

PROFESSIONAL EXPERIENCE

AI Engineer <i>Betopia LTD</i>	Jul 2025 – Present <i>Dhaka, Bangladesh</i>
- Designed and deployed production-grade AI applications using OpenAI, Claude, Gemini, and self-hosted LLMs - Built Retrieval-Augmented Generation (RAG) systems using vector databases including Qdrant and PGVector to improve knowledge retrieval and contextual reasoning - Developed AI agents and coding assistants leveraging LangChain-based workflows to automate engineering and operational tasks - Architected a centralized LLM platform providing unified API access to multiple commercial and open-source model providers - Built FastAPI-based backend services and inference APIs supporting scalable AI applications - Containerized AI services using Docker and optimized deployment workflows for production environments - Fine-tuned transformer-based models for domain-specific use cases using proprietary datasets - Improved inference performance through model serving optimizations, intelligent routing, and infrastructure enhancements - Collaborated with engineering and business stakeholders to translate requirements into AI-powered solutions	

EDUCATION

B.Sc. in Computer Science and Engineering <i>North South University</i>	2025 <i>Dhaka, Bangladesh</i>
Professional Development: Short Course on Data Science — Prof. Jennifer Widom, Stanford University <i>Feb 2024</i>	
Certifications: Building with the Claude API · Claude Code in Action · Introduction to Agent Skills · Introduction to Model Context Protocol <i>Anthropic, May 2026</i>	

KEY PROJECTS

Betopia AI Ecosystem	<i>Go, Python, TypeScript, FastAPI, vLLM, PostgreSQL, Redis</i>
- Architected and developed a production AI ecosystem consisting of conversational AI, coding agents, and developer-facing inference APIs - Implemented Retrieval-Augmented Generation (RAG) pipelines with semantic search and vector retrieval capabilities	

- Built multi-agent workflows supporting code generation, task execution, workflow automation, and tool usage
- Developed scalable FastAPI and Go services supporting high-volume LLM inference and AI application workloads
- Integrated OpenAI, Claude, Gemini, Groq, and self-hosted models through a unified orchestration layer
- Designed authentication, rate limiting, monitoring, RBAC, and provider failover mechanisms for production reliability
- Contributed as a founding AI team member in building the company's flagship AI platform

AI-Based Grocery Checkout System

Python, OpenCV, YOLOv8, RT-DETR

- Built an AI-powered checkout system capable of detecting and classifying grocery products from a single image
- Trained and evaluated YOLOv8 models on a custom dataset, achieving **0.995 mAP@50**
- Optimized inference performance to **696 ms/frame** enabling near real-time operation
- Designed product identification workflows using GTIN-based product mapping for automated checkout simulation

Enterprise RAG Knowledge Assistant

Python, LangChain, Qdrant, FastAPI, OpenAI

- Developed a Retrieval-Augmented Generation (RAG) assistant for enterprise knowledge retrieval and question answering
- Built document ingestion, chunking, embedding generation, vector indexing, and retrieval pipelines
- Integrated OpenAI models with vector search to provide context-aware responses grounded in organizational knowledge
- Implemented FastAPI endpoints and scalable retrieval workflows for low-latency user interactions